

Task 2

Title

Weather API Integration using Python

Objective

To fetch live weather data using API and process JSON response.

Tools Used

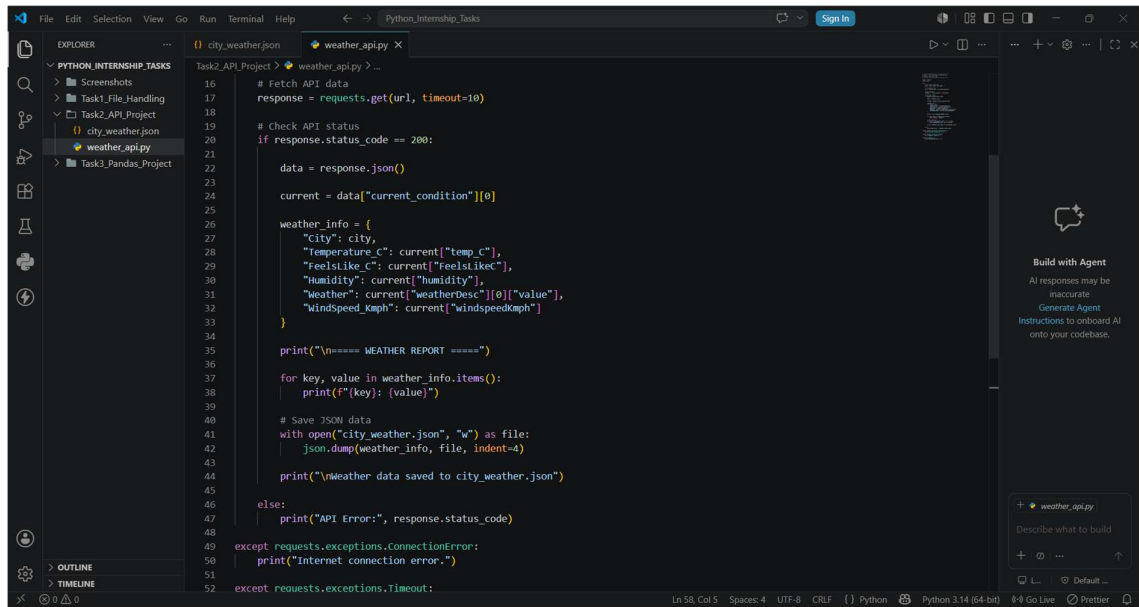
- Python
- Requests Library
- JSON Module

Concepts Used

- API Integration
- JSON Parsing
- Exception Handling
- Conditional Logic

Screenshot

- Screenshot1



The screenshot displays a code editor with a dark theme. The Explorer panel on the left shows a project structure with folders for 'PYTHON_INTERNSHIP_TASKS', 'Task1_File_Handling', 'Task2_API_Project', and 'Task3_Pandas_Project'. Inside 'Task2_API_Project', there are files 'city_weather.json' and 'weather_api.py'. The 'weather_api.py' file is open in the editor, showing a Python script that fetches weather data from an API, parses the JSON response, and saves it to a file. The script includes comments and uses the 'requests' and 'json' modules. It also features exception handling for 'ConnectionError' and 'Timeout'. The status bar at the bottom indicates the file is at line 58, column 5, using UTF-8 encoding with CRLF line endings, and is running on Python 3.14 (64-bit).

```
16 # Fetch API data
17 response = requests.get(url, timeout=10)
18
19 # Check API status
20 if response.status_code == 200:
21
22     data = response.json()
23
24     current = data["current_condition"][0]
25
26     weather_info = {
27         "City": city,
28         "Temperature_C": current["temp_c"],
29         "Feelslike_C": current["feelslike_c"],
30         "Humidity": current["humidity"],
31         "Weather": current["weatherDesc"][0]["value"],
32         "WindSpeed_Kmph": current["windspeedkmph"]
33     }
34
35     print("\n==== WEATHER REPORT =====")
36
37     for key, value in weather_info.items():
38         print(f"{key}: {value}")
39
40     # Save JSON data
41     with open("city_weather.json", "w") as file:
42         json.dump(weather_info, file, indent=4)
43
44     print("\nWeather data saved to city_weather.json")
45
46 else:
47     print("API Error:", response.status_code)
48
49 except requests.exceptions.ConnectionError:
50     print("Internet connection error.")
51
52 except requests.exceptions.Timeout:
```

